Senthen Velmurugan

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Education

Georgia Institute of Technology **In Progress**
Master of Science in Computer Science *Atlanta, GA*

The University of Texas at Dallas
Bachelor of Science in Computer Science *Dallas, TX*

Experience

KloudGin **May 2024 – August 2024**
Machine Learning Engineering Intern *Remote*

- Took the Generative AI team's open question, whether RAG could speed up customer case resolution, from proposal to production rollout, raising first-contact resolution 20% over 12 weeks of use by support engineers
- Built a multi-source RAG system that retrieves similar past cases from the ticketing corpus and accepted solutions from the internal forum, then suggests next actions, with an ingestion pipeline (normalization, chunking, metadata tagging), hybrid vector + BM25 retrieval and cross-encoder re-ranking
- Measured it with offline retrieval metrics and an A/B feedback loop with support agents; when it overfit to long threads, retuned chunking and metadata filters to raise top-k precision
- Added tenant-scoped search, PII scrubbing and safe prompting, and exposed it through a documented API for other teams to integrate

GBC Computers **May 2023 – August 2023**
Cybersecurity and Systems Intern *Mabank, TX*

- Improved client network performance by monitoring with PRTG and tracing bottlenecks in Wireshark captures
- Hardened client infrastructure with pfSense firewall rules, Snort intrusion detection, and endpoint protection
- Found and patched critical risks with Nessus vulnerability assessments, validated by OpenVAS and manual audits

Projects

Campbell Woodworks: Storefront for a Family Woodshop | *Jekyll, HTML, CSS, Python* **2026**

- Delivered a live storefront for a family woodshop whose work lived only in Facebook posts behind a login, so buyers can browse between farmers-market days with nothing for the shop to run or pay for
- Built the site around how they sell (by phone, not checkout): one Markdown file per piece drives the shop, gallery and detail pages, with category filters and call and email buttons, covering 23 pieces

P&ID Digitizer: Scanned Engineering Drawings to Graphs | *Python, PyTorch, OpenCV, FastAPI, Docker* **2026**

- Turned raster piping and instrumentation diagrams, the drawings that hold a plant's process logic, into queryable graphs (GraphML, DEXPI 2.0.1 XML) at 0.915 edge F1 and 0.997 symbol mAP@0.5 on a 75-sheet test split
- Detected small symbols on 7,000 px sheets by tiling into overlapping 1024^2 crops with class-aware box merging across seams, fine-tuning a Faster R-CNN in under 2 GPU-hours on a laptop GPU
- Shipped it as a FastAPI job service in a non-root Docker image, with calibrated per-edge confidence feeding a review UI that sends only doubtful edges to a person; strict mypy typing and 99 pytest cases in CI

DEXPI P&ID Viewer | *JavaScript, Python, XML, SVG* **2026**

- Built a zero-dependency, single-page viewer and validator for DEXPI 2.0.1 plant models that lays out diagrams with no stored coordinates from topology alone, and checks structure and the Core and Plant metamodel in the browser
- Gave data pipelines a headless path with a Python CLI validator (lxml) running both XSD and metamodel checks

Word2Eq: Reproducible Seq2Seq Training & Inference Platform | *Python, PyTorch, FastAPI, Prometheus* **2026**

- Rebuilt a non-functional prototype into a train/eval/serve platform for a 6.4M-parameter Transformer; grammar-constrained decoding took valid equations from 97.4% to 100% on 1,000 SVAMP problems
- Served 150 requests/s at 73 ms p95 on one laptop CPU by micro-batching behind a bounded queue that sheds overload with a fast 503, and made training preemption-safe with bit-identical resumes

Technical Skills

Languages: Python, JavaScript, Go, C/C++, SQL, Bash
Frameworks: PyTorch, FastAPI, LangChain, OpenCV, pandas
Tools: Linux (Debian), Git, Docker, Kubernetes, Terraform, PostgreSQL, Pinecone, CI/CD, Autodesk Fusion

Organizations

Mensa **2013 – Present**